Invariant Patterns under "Kasami is Almost Bent"

Tiny atoms that propagate + which ones warrant a categorical reading

Abstract

We collect the *smallest* reusable lemmas under the Lean proof "Kasami is Almost Bent" and show they are all one idea wearing six costumes:

$$\boxed{\text{property } P \text{ is unchanged when we move along a symmetry}}$$

In type theory / category theory this is *transport along an iso*: P is a **fixed point** of an automorphism action, i.e. an *invariant*.

We give each atom a formula, a picture, and the real Lean snippet (file `Kasami/InvariantPatterns.lean`, fully proved, no `sorry`). Then we read `FiniteField/Thm32.lean` and sort *every* supporting lemma into **functorial** vs **algebra / Galois / reindexing** — to see which actually deserve a categorical reading.

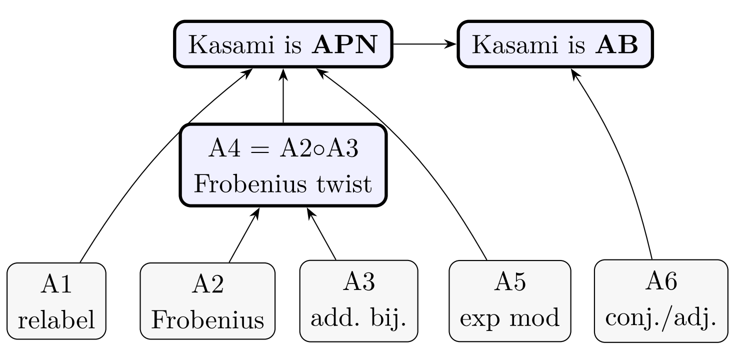
★ BIG IDEA

One symmetry $\Rightarrow$ one invariance lemma. Stack the lemmas $\Rightarrow$ the headline theorem.

$$\underbrace{\text{Mathlib atoms}}_{\text{leaves}} \xrightarrow{\text{transport}} \underbrace{\text{Kasami is APN}}_{\text{pillar}} \xrightarrow{\text{transport}} \underbrace{\text{Kasami is AB}}_{\text{crown}}$$

1 The shape: six atoms, one law

A *symmetry* is an iso σ. An *invariance lemma* says: $P(\sigma \cdot x) \iff P(x)$.



✎ REMARK

Read bottom → top. Each arrow is "transport along a symmetry". The *same* five atoms reappear in `Lemma31`, `FrobAlg`, `KasamiEvenK`.

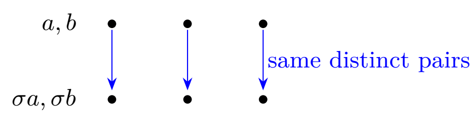
2 Atom 1 — relabel a parameter

Symmetry: a bijection σ on the index set.

$$\big(\forall a \neq b,\ R(\sigma a, \sigma b)\big) \iff \big(\forall a \neq b,\ R(a, b)\big)$$

A "for all distinct pairs" fact does not care how we *name* the pairs.

```
lemma forall_pairs_relabel α{ : Type∗} σ{ : α → α} (σh : Bijective σ)
    {R : α → α → Prop} :
    ∀( a b, a ≠ b → R σ( a) σ( b)) ↔ ∀( a b, a ≠ b → R a b)
```



✦ COOL FACT

This is **naturality** of the predicate ∀-pairs along σ. In `Lemma31` it slides the bijectivity test along the multiplicative map M (`forall_ne_bij`).

3 Atom 2 — Frobenius is an additive automorphism

Symmetry: $\phi_j(x) = x^{2^j}$ on GF(2^n).

$$\phi_j(x + y) = \phi_j x + \phi_j y \qquad \text{and} \qquad \phi_j \in \mathrm{Aut}(F, +).$$

Freshman's dream in char 2 ⇒ additive; finite + injective ⇒ bijective.

```
lemma frob_add (j : ℕ) (x y : F) :
    (x + y) ^ (2 ^ j) = x ^ (2 ^ j) + y ^ (2 ^ j)

lemma frob_bij (j : ℕ) : Bijective (fun x : F => x ^ (2 ^ j))
```

$$
\begin{tikzcd}
F \arrow[r, "\phi_j"] \arrow[d, "+y"'] & F \arrow[d, "+\phi_j y"] \\
F \arrow[r, "\phi_j"'] & F
\end{tikzcd}
$$

✎ REMARK

Lean fact used: `add_pow_char_pow` (Mathlib) and `Finite.injective_iff_bijective`. This is the *only* automorphism the APN pillar needs.

4 Atom 3 — APN survives an additive bijection

Symmetry: post-compose by an additive bijection σ.

$$\mathrm{IsAPN}(f) \implies \mathrm{IsAPN}(\sigma \circ f).$$

APN counts collisions of $f(x+a) + f(x)$. An additive *bijection* keeps collisions exactly, so the count is unchanged.

```lean
lemma isAPN_comp_addBij {f σ : F → F} (hf : IsAPN f)
    (hbij : Bijective σ) (hadd : ∀ x y, σ (x + y) = σ x + σ y) :
    IsAPN σ( ∘ f)
```

$$
\begin{tikzcd}
x \arrow[r, "\text{2-to-1 diff}"] \arrow[d, "f"'] & b \\
F \arrow[r, "\sigma\ (\cong)"'] & F \arrow[u, "\text{same count}"']
\end{tikzcd}
$$

✦ COOL FACT

This is *the* propagation pattern the user spotted: "additive bijection" travels from `FrobAlg`/Mathlib into `KasamiEvenK`. APN is a **conjugation-invariant** of the additive group.

5 Atom 4 — Frobenius exponent twist (A2 ∘ A3)

Glue Atom 2 into Atom 3:

$$x^{d\cdot 2^j} = (x^d)^{2^j} = \phi_j(x^d) \qquad \Rightarrow \qquad \mathrm{IsAPN}(x^d) \Rightarrow \mathrm{IsAPN}(x^{d\cdot 2^j}).$$

```
lemma isAPN_frob_twist {d : ℕ} (hf : IsAPN (fun x : F => x ^ d)) (j : ℕ) :
    IsAPN (fun x : F => x ^ (d * 2 ^ j))
```

$$\begin{tikzcd} & x \arrow[dl, "x^d"'] \arrow[dr, "x^{d\cdot 2^j}"] & \\ x^d \arrow[rr, "\phi_j"'] & & x^{d\cdot 2^j} \end{tikzcd}$$

★ BIG IDEA

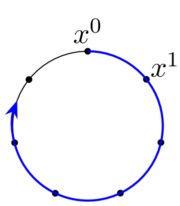
A4 is the engine of `kasami_apn_of_complement`: it lifts Kasami-APN from a parameter k to its complement $n - k$, with *no parity restriction*. APN is invariant along the **whole Frobenius orbit** of the exponent.

6 Atom 5 — slide the exponent mod $|F| - 1$

Symmetry: $\mathbb{Z}/(|F|-1)$ acting on exponents (for $x \neq 0$).

$$a \equiv b \pmod{|F| - 1} \implies x^a = x^b.$$

```
lemma pow_invariant_mod {x : F} (hx : x ≠ 0) {a b : ℕ}
    (h : a % (Fintype.card F - 1) = b % (Fintype.card F - 1)) :
    x ^ a = x ^ b
```



exponents live on a circle of length $|F| - 1$

✎ REMARK

This is the wrap-around behind the Kasami congruence $d_k \equiv d_{n-k} \cdot 2^{2k} \pmod{2^n - 1}$ in `KasamiEvenK`.

7 Atom 6 — bijectivity is a conjugation invariant

Symmetry: conjugate a self-map by an equivalence e.

$$\mathrm{Bijective}(e \circ f \circ e^{-1}) \iff \mathrm{Bijective}(f).$$

```
lemma bijective_conj_invariant α{ β : Type*} (e : α ≃ β) (f : α → α) :
    Bijective (e ∘ f ∘ e.symm) ↔ Bijective f
```

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "e"', "\cong"] & A \arrow[d, "\cong"', "e"] \\
B \arrow[r, "efe^{-1}"'] & B
\end{tikzcd}$$

✦ COOL FACT

The trace-adjoint $A \mapsto A^*$ in `Lemma31` (`bijective_iff_adjoint_bijective`) is exactly such a symmetry: **duality** keeps bijectivity. “Is it a permutation?” is a self-dual question.

8 How the atoms reach the crown

$$\begin{tikzcd}
\mathrm{A2} + \mathrm{A3} \arrow[r, Rightarrow] & \mathrm{A4} \arrow[dr, Rightarrow] & \\
\mathrm{A5} \arrow[r, Rightarrow] & \mathrm{A4 \ on \ } d_k \arrow[r, Rightarrow] & \boxed{\text{Kasami APN, all } k} \arrow[d, Rightarrow] \\
 & & \boxed{\textbf{Kasami AB}}
\end{tikzcd}$$

The Lean synthesis is one line:

```
theorem isAPN_invariant_along_frobenius_orbit {d : ℕ}
    (hf : IsAPN (fun x : F => x ^ d)) (j : ℕ) :
    IsAPN (fun x : F => x ^ (d * 2 ^ j)) :=
  isAPN_frob_twist hf j
```

9 Reading `Thm32.lean`: functorial vs algebra / Galois / reindexing

Theorem 3.2 (Dempwolff–Müller) proves $x \mapsto L(x)\, x^k$ is a permutation of $\mathrm{GF}(2^n)$, with $L(x) = \sum_{i<m} x^{2^i}$ the truncated trace.

We sort *every* supporting lemma into four bins. Only the first bin warrants a categorical reading.

lemma	bin	why
`truncTrace_add`, `_zero`	**functorial**	L is a morphism in **Ab** (`map_add`, `map_zero`)
`dicksonF_map_ringHom`	**functorial**	naturality: f_m commutes with *every* ring hom (`map_sum`, `map_pow`)
`LxXk'_bijective` (adjoint swap)	**functorial**	trace-duality / adjunction (`adjoint_swap_bij`, `trace_nondegenerate`)
`LxXk_bijective` (inj. part)	**functorial**	finite: inj $\Leftrightarrow$ surj (`Finite.injective_iff_surjective`)
`truncTrace_sq_add_self`	algebra	Artin–Schreier identity $L^2{+}L = x^{2^m}{+}x$
`truncTrace_one_eq_one`	algebra	char-2 parity count
`sq_eq_self_imp`	algebra	root count $x^2{=}x \Rightarrow x \in \{0,1\}$
`dicksonF_one`, `_recursion_mul`	algebra	polynomial recursion
`dicksonF_functional`	algebra	$f_m(z{+}z^{-1})$ functional eq.
`eq_or_eq_inv_of_add_inv_eq`	algebra	fibres of $z \mapsto z{+}z^{-1}$
`truncTrace_sq_mul_inv_eq_dicksonF`	algebra	rearrange a finite sum
`frob_fixed_of_truncTrace_zero`	Galois	$\ker L \subseteq \mathrm{Fix}\,\phi_m$
`truncTrace_ker_trivial`	Galois	fixed field of gcd Frobenius
`exists_add_inv_rep`	Galois	base change to $\overline{F}$
`frob_2n_eq_self_of_quad_root`	Galois	Frobenius descent of a root
`truncTrace_adj_frob`	Galois	Frobenius reindex of the trace
`coprime_mersenne_double'`	reindexing	$\gcd(2^a{-}1, 2^b{-}1) = 2^{\gcd(a,b)}{-}1$
exponent mod $2^n{-}1$ slides	reindexing	$\mathbb{Z}/(2^n{-}1)$ arithmetic

★ BIG IDEA

Verdict. Out of ∼22 lemmas, only **four shapes** are genuinely functorial:

$$\underbrace{L\text{ additive}}_{\text{morphism in }\mathbf{Ab}}\;,\quad \underbrace{f_m \circ \mathrm{hom} = \mathrm{hom} \circ f_m}_{\text{natural transf.}}\;,\quad \underbrace{A \leftrightarrow A^*}_{\text{adjunction}}\;,\quad \underbrace{\text{inj} \Leftrightarrow \text{surj}}_{\text{finite-set invariant}}\;.$$

Everything else is char-2 **algebra**, **Galois** descent, or modular **reindexing** — field-specific computation, *not* category theory.

✦ COOL FACT

The four functorial lemmas are exactly the *invariant atoms* of Part I:

$$\text{additivity} \sim \text{A2/A3}, \qquad \text{naturality} \sim \text{A1}, \qquad \text{adjoint} \sim \text{A6}.$$

So "which lemmas warrant a categorical reading?" has the same answer as "which atoms are transport-along-an-iso?" — they are the load-bearing **invariance** steps, while the algebra/Galois bulk supplies the one hard input (a single map is a bijection).

10 One sentence to remember

☞ *Six symmetries, six invariance lemmas, one crown — and only the* ***transport*** *steps are categorical.*